

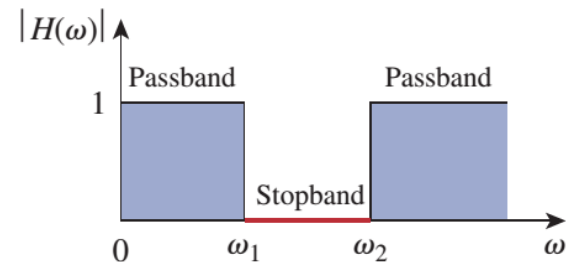
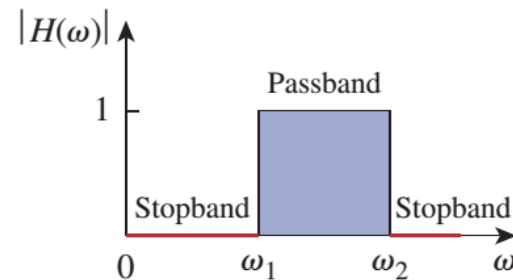
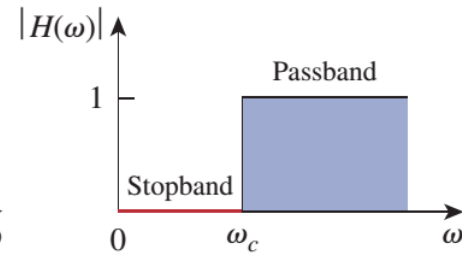
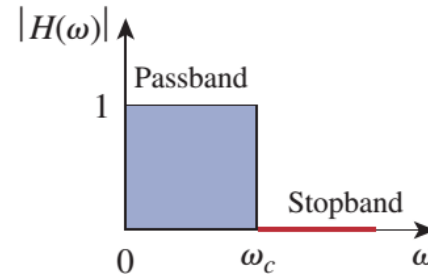


Lecture 14

- Filters

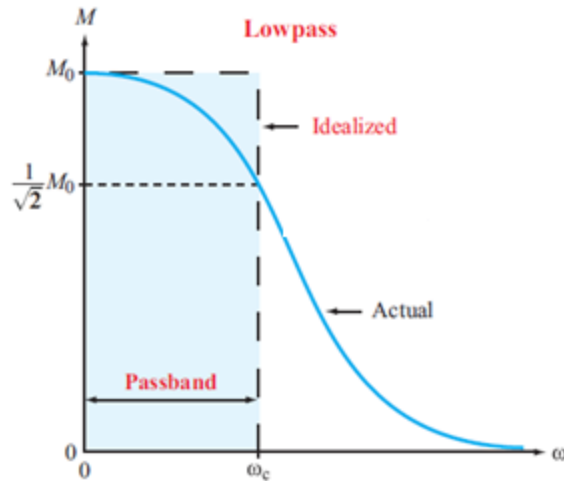
Filters

- Circuit designed to retain certain frequency range but discard or attenuate others
 - *Low-pass*: pass low frequencies and reject high frequencies
 - *High-pass*: pass high frequencies and reject low frequencies
 - *Band-pass*: pass some particular range of frequencies, reject other frequencies outside that band
 - *Band-reject (notch)*: reject a range of frequencies and pass all other frequencies

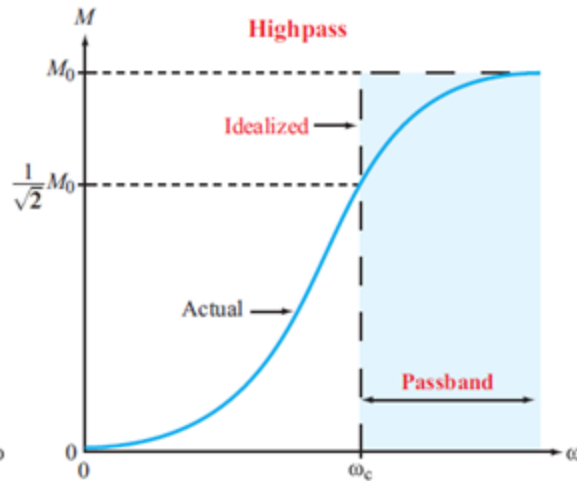




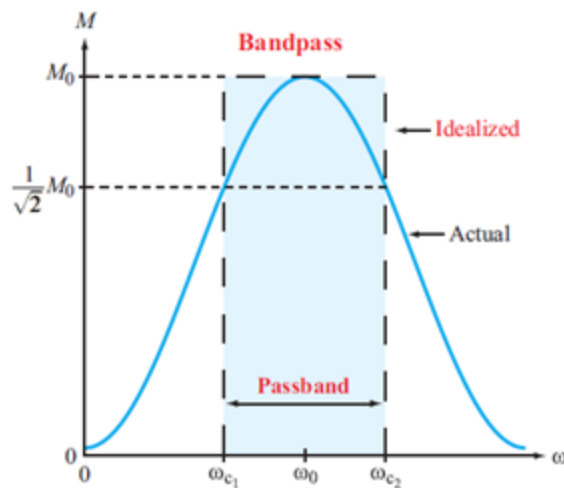
Filters – Realistic Curves



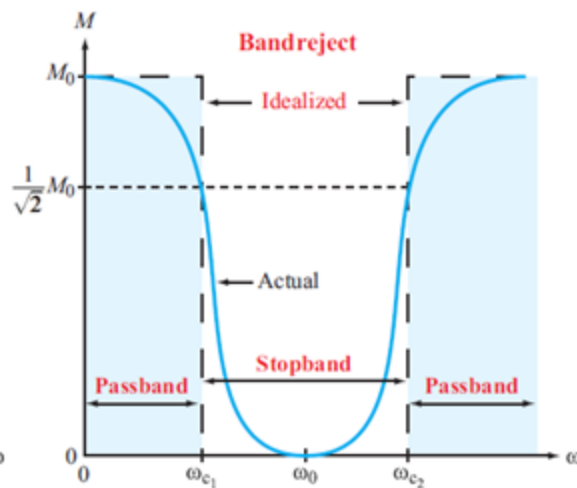
(a) Lowpass filter



(b) Highpass filter



(c) Bandpass filter



(d) Bandreject filter

[Source: Berkeley]



Passive Filters

- A filter is passive if it consists only of passive elements
 - R, L and C
- LC filters have been used in practical applications for decades
 - Very important circuits
 - many technological advances would not have been possible without the development of filters
 - LC filter technology feeds many areas
 - equalizers, impedance-matching networks, transformers, shaping networks, power dividers, attenuators, and directional couplers ...

First-Order RC Lowpass Filter

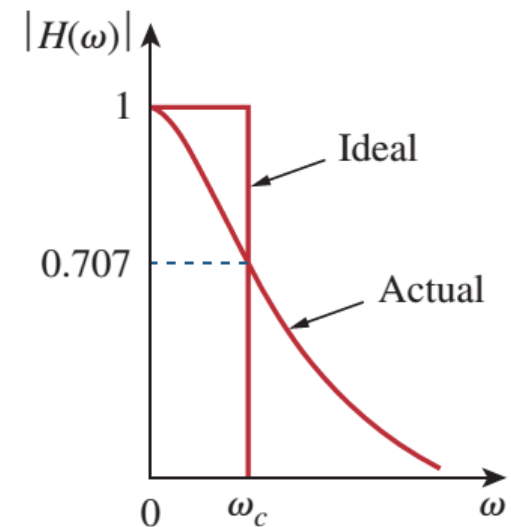
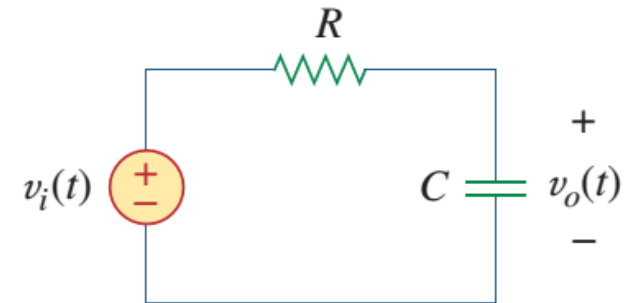
- A typical lowpass filter is formed when the output of a RC circuit is taken off the capacitor.

$$\mathbf{H}(\omega) = \frac{1}{1 + j\omega RC}$$

- The -3dB (half power) frequency is:

$$\omega_c = \frac{1}{RC}$$

- Also referred as the cutoff frequency.
- Filter is designed to pass from DC up to ω_c .



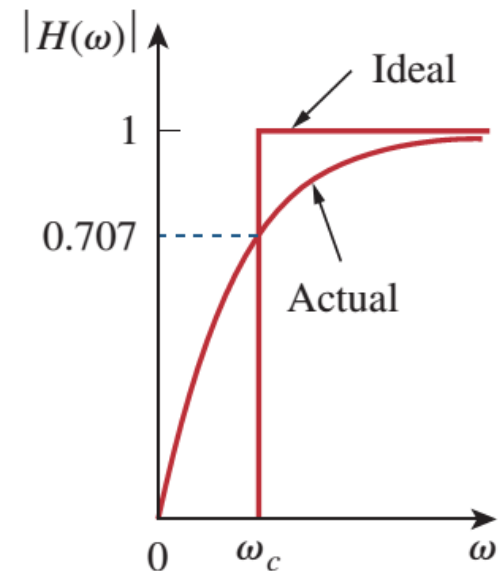
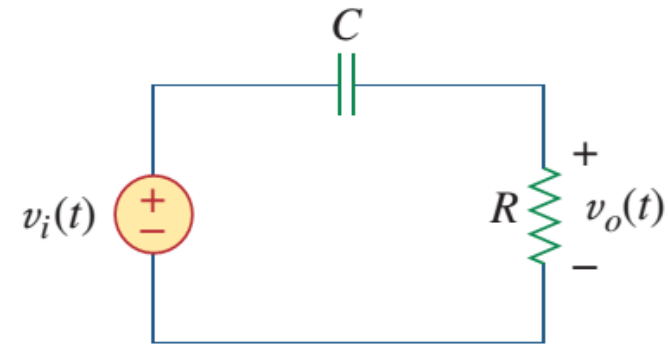
First-Order RC Highpass Filter

$$\mathbf{H}(\omega) = \frac{1}{1 + \frac{1}{j\omega RC}}$$

- The cutoff frequency will be the same as the lowpass filter.

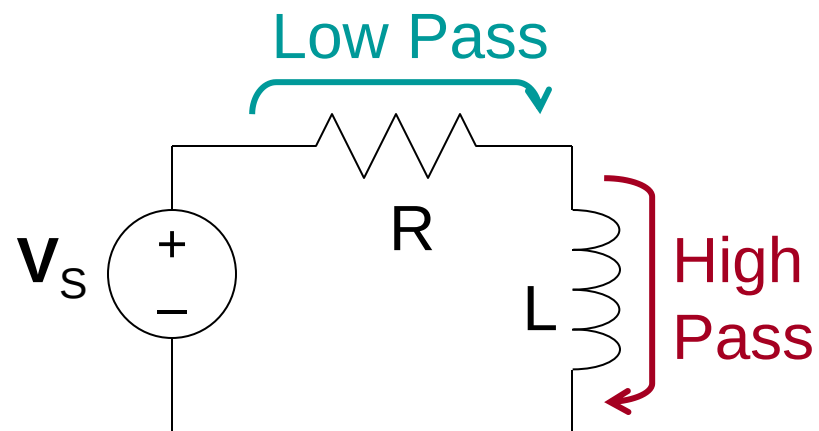
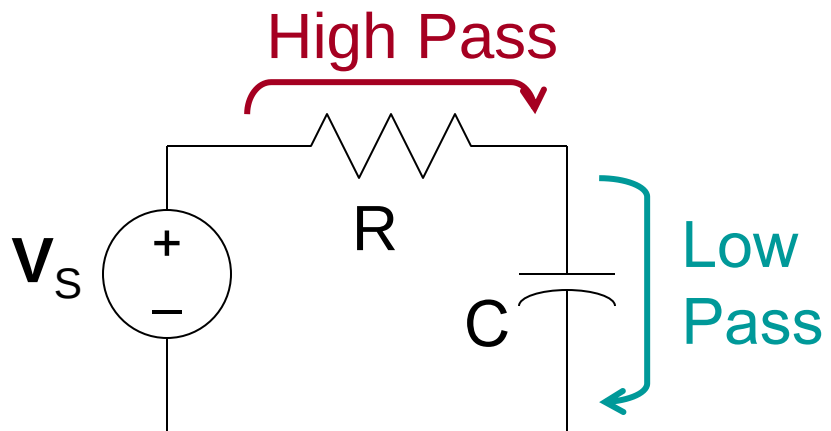
$$\omega_c = \frac{1}{RC}$$

- The difference being that the frequencies passed go from ω_c to infinity.



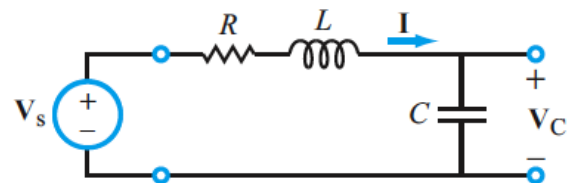
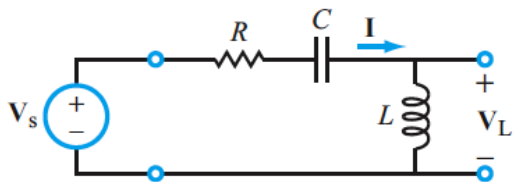


How about RL Circuits?



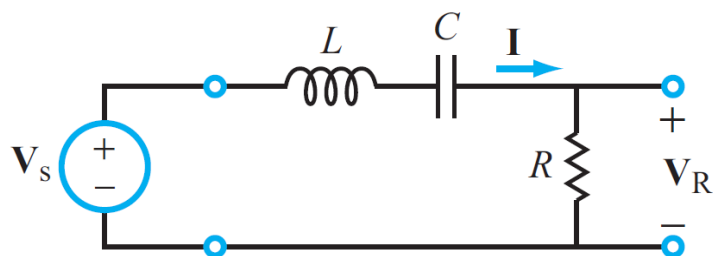


How about RLC Circuits?

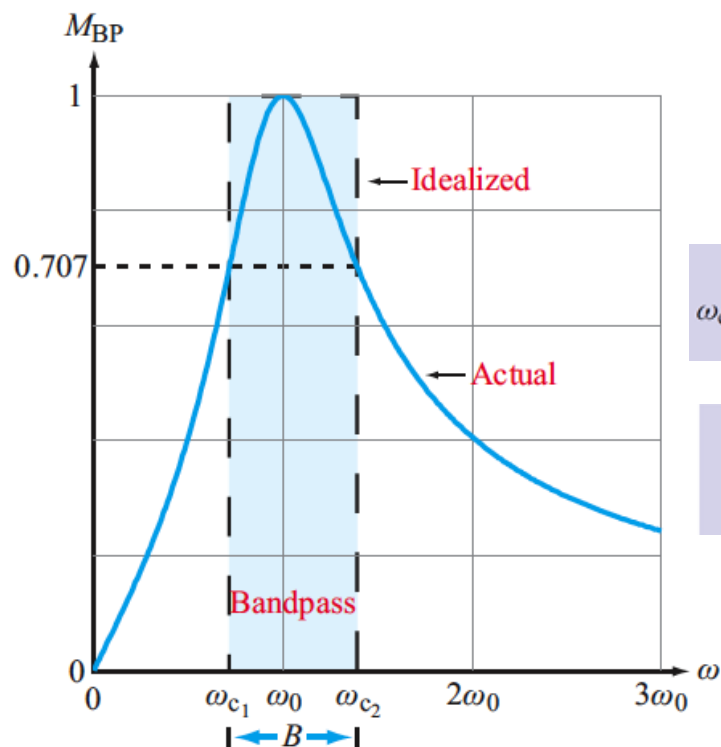




Bandpass RLC Filter



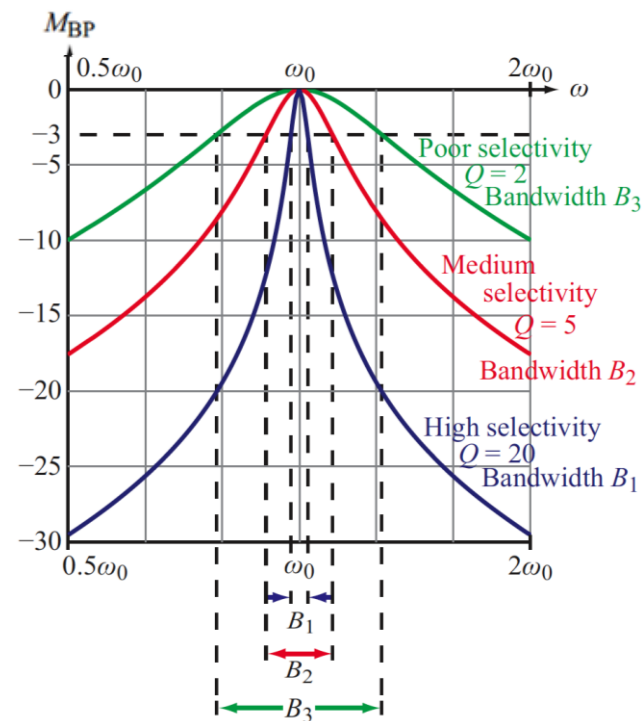
$$\mathbf{H}_{BP}(\omega) = \frac{\mathbf{V}_R}{\mathbf{V}_s} = \frac{R}{Z}$$



$$\omega_0 = \frac{1}{\sqrt{LC}}$$

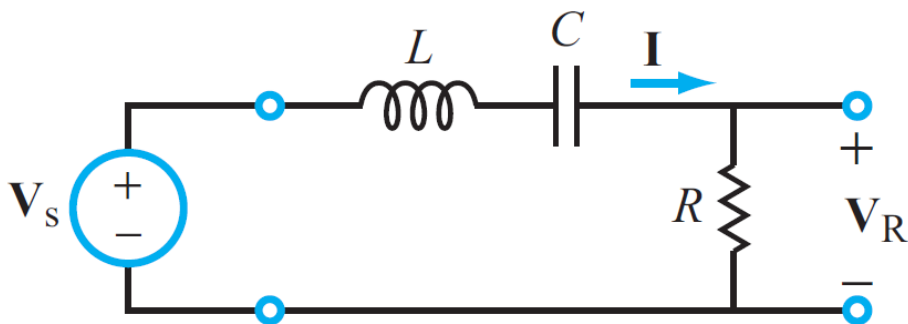
$$\omega_{c1} = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}},$$

$$\omega_{c2} = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}.$$

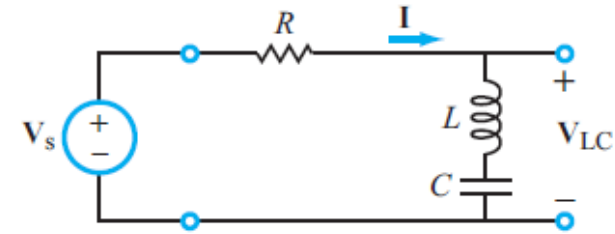


Example

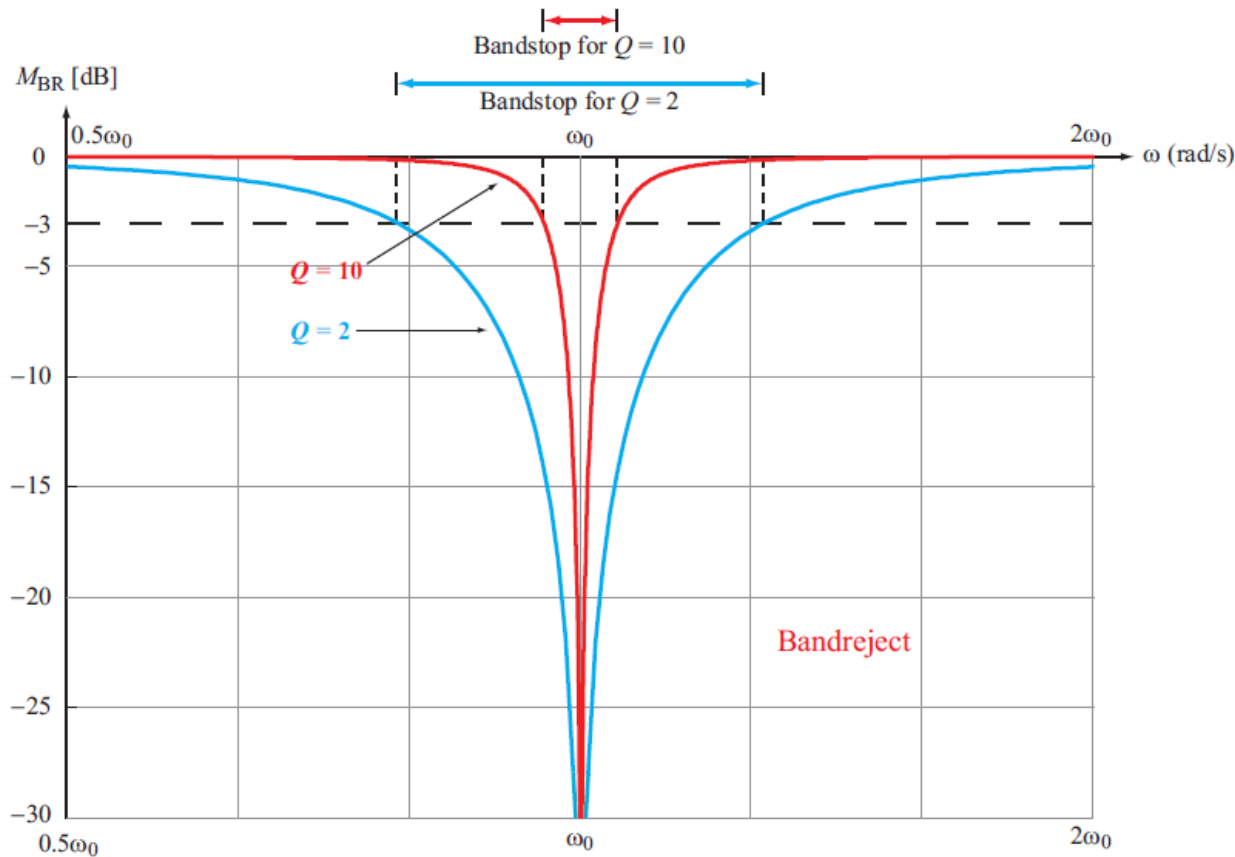
- (a) Design a series RLC bandpass filter with a center frequency $f_0 = 1$ MHz and a quality factor $Q = 20$, given that $L = 0.1$ mH.



Bandstop (Bandreject) Filter



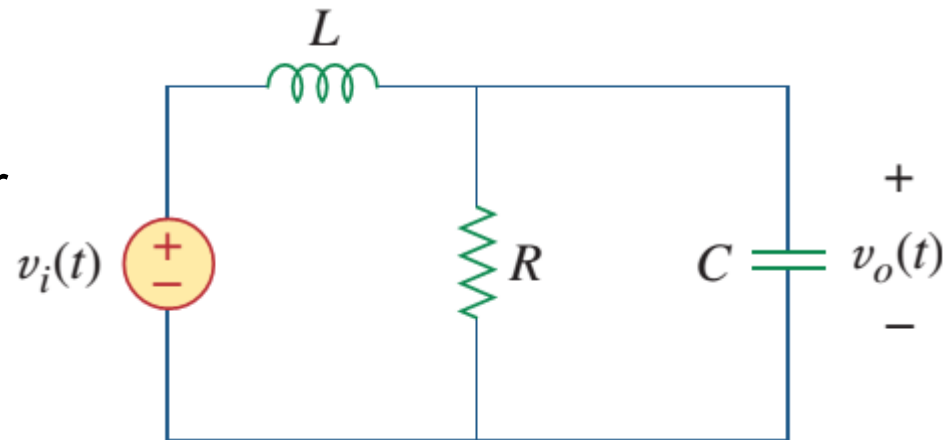
$$H_{BR} = V_{LC} / V_s$$



Example

- Determine what type of filter is shown below. Calculate the corner or cutoff frequency.

$R = 2\text{k}\Omega$, $L = 2\text{H}$ and $C = 2\mu\text{F}$.

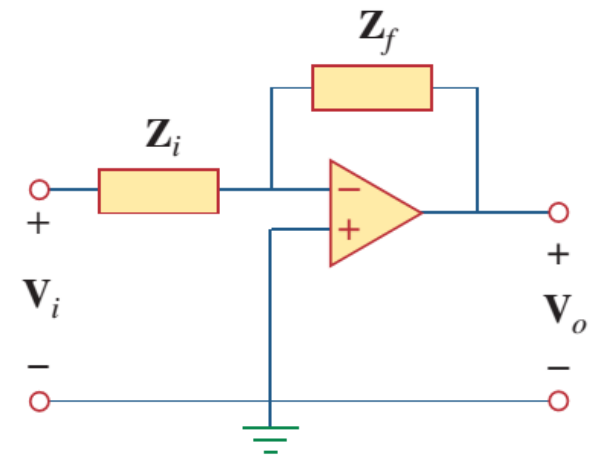
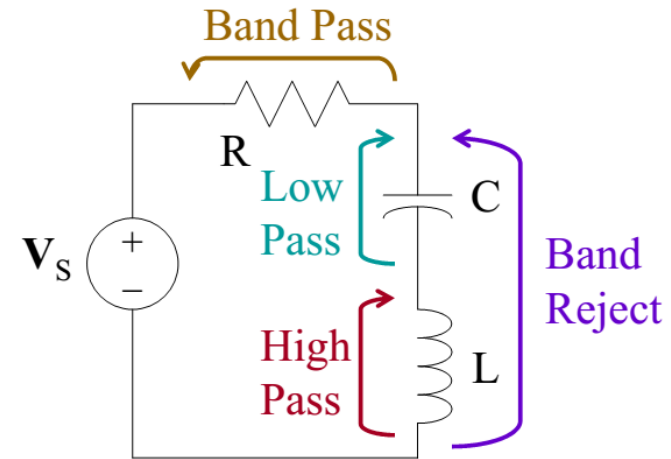


second-order lowpass filter

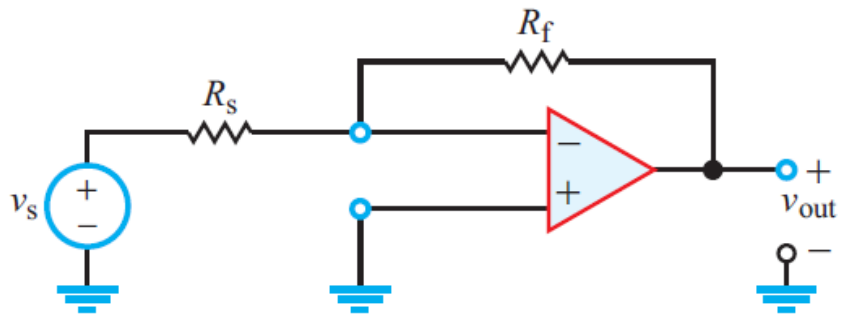
$$\omega_c = 0.742 \text{ krad/s} = 742 \text{ rad/s}$$

Active Filters

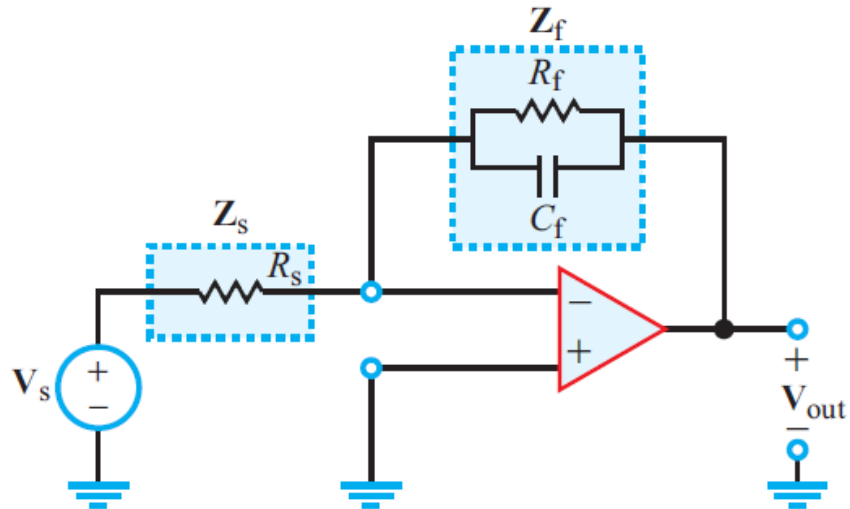
- Passive filters have a few drawbacks.
 - Gain.
 - Impedance.
- It is possible, using op-amps, together with resistors and capacitors, to create all the common filters.
 - Their ability to isolate input and output also makes them very desirable.
 - Limited to frequency less than 1MHz.



Active Filters – Lowpass



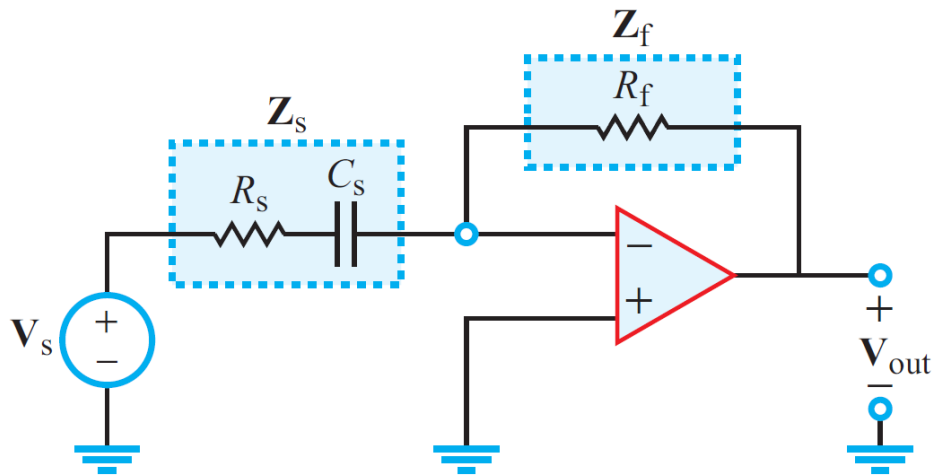
(a) Inverting amplifier



(b) Phasor domain with impedances

$$H(\omega) = -\frac{Z_f}{Z_s}$$

Active Filters – Highpass



$$\mathbf{H}_{HP}(\omega) = \frac{V_{out}}{V_s} =$$

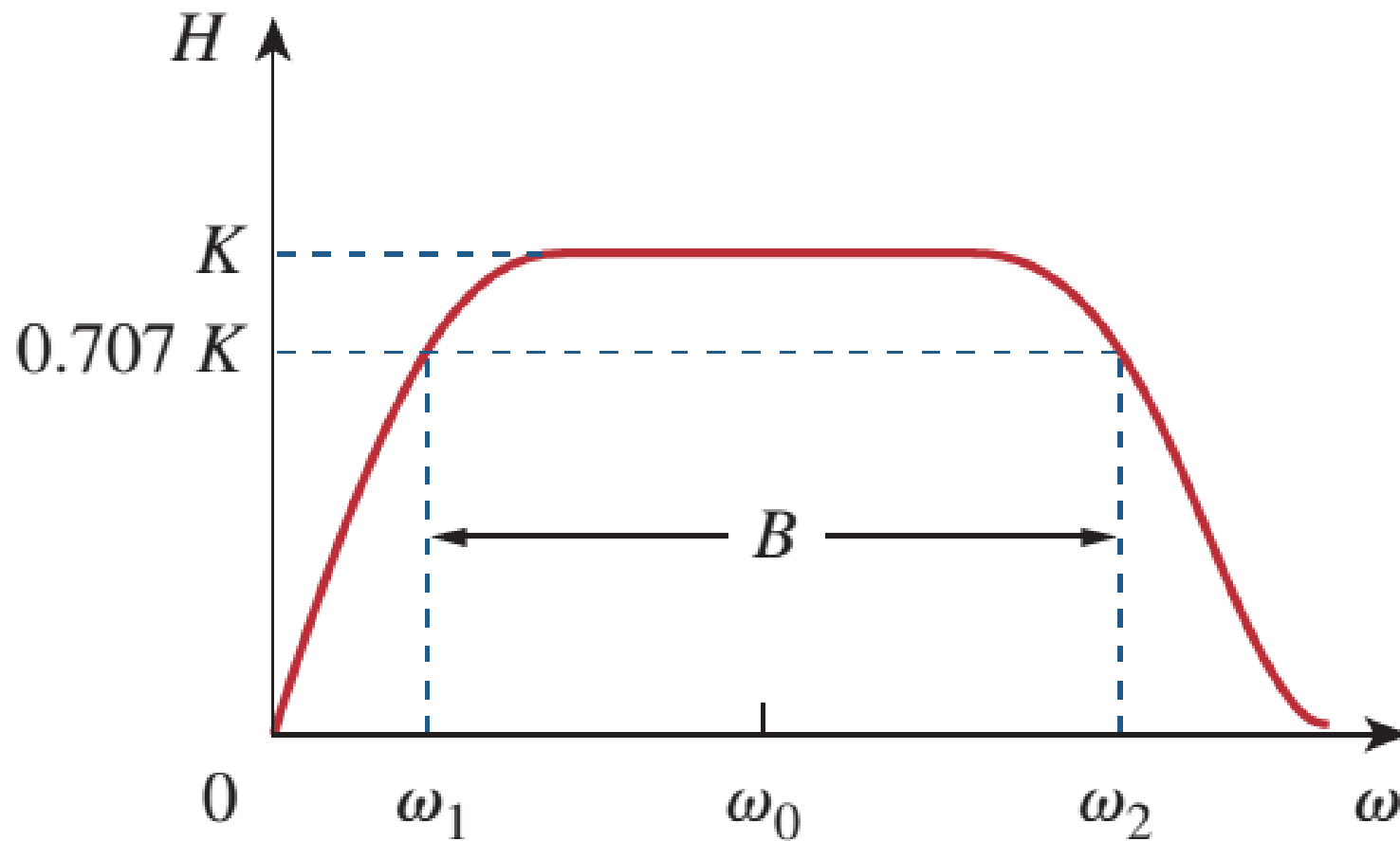


Exercise

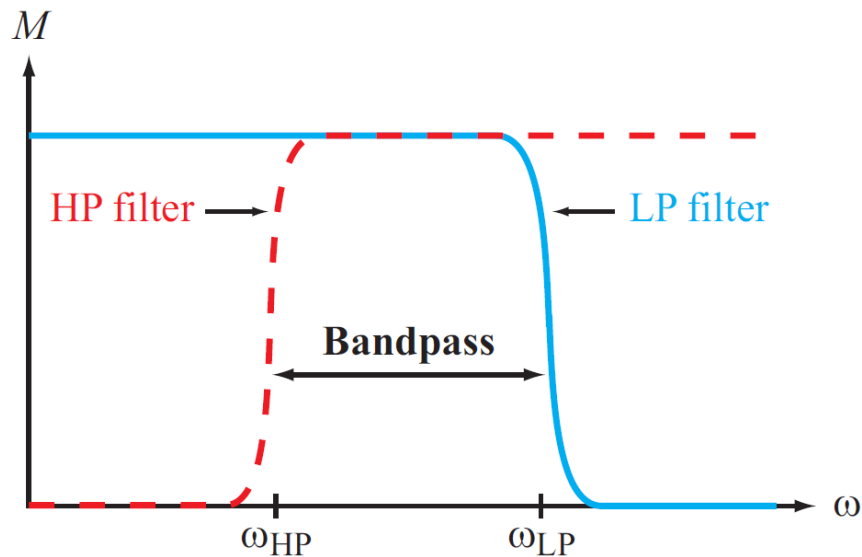
- Design a 1st order low-pass filter having a gain of 1 in the passband and a cutoff frequency 1 rad/s.



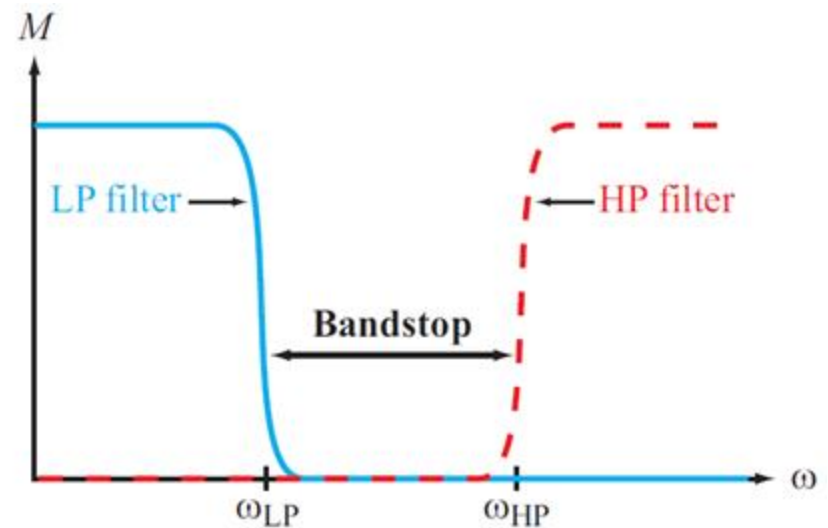
Bandpass



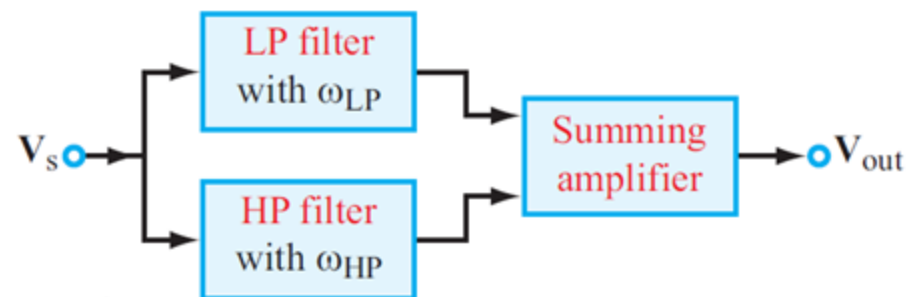
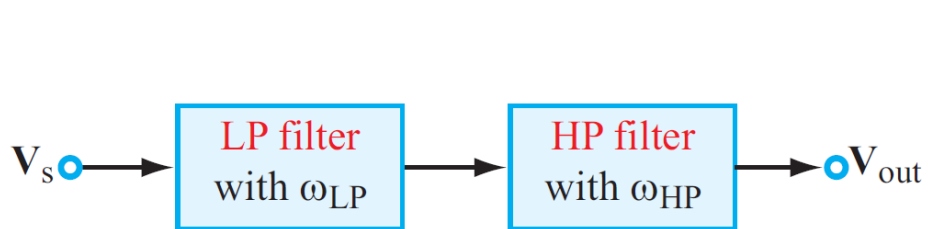
Cascading Active Filters



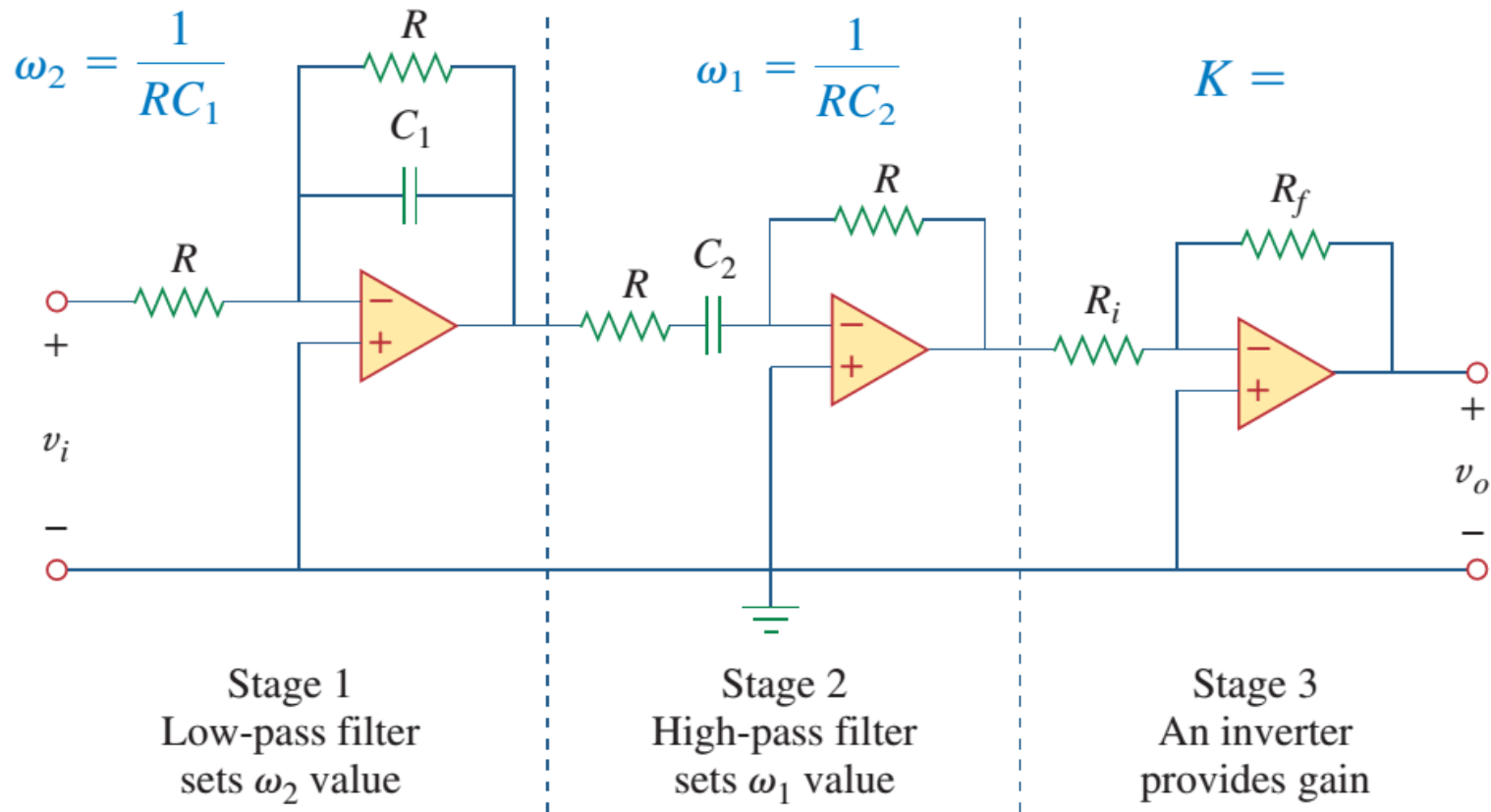
(a) Bandpass filter



(b) Bandreject filter



Active Bandpass Filter



$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \left(-\frac{1}{1 + j\omega C_1 R} \right) \left(-\frac{j\omega C_2 R}{1 + j\omega C_2 R} \right) \left(-\frac{R_f}{R_i} \right)$$

Active Bandreject Filter

